Artificial Intelligence

Unit 1

Artificial Intelligence

Artificial Intelligence (AI): AI is the study of making machines to perform tasks that require human intelligence.

How AI works: AI systems process large amounts of data and apply algorithms (smart rules, models, or techniques) to recognize patterns, learn from experiences, and make accurate decisions or predictions.

Importance of AI: AI is valuable because it can solve complex problems, increase efficiency by completing tasks faster, reduce human effort in repetitive or risky work, and provide support in fields like healthcare, education, business, and technology.

Applications of AI

Applications of AI in Healthcare: Automated disease identification and diagnosis, telemedicine and virtual health assistants, robot-assisted surgery, healthcare chatbots.

Applications of AI in Automobile Industry: Autonomous vehicles (self-driving cars), traffic prediction and routing, infotainment systems, voice recognition and natural language processing, driver assistance for parking.

Applications of AI in Financial Industry: Trading, fraud detection and prevention, portfolio management and optimization, credit scoring, cybersecurity.

Applications of AI in Surveillance and Security: Crowd analysis, traffic management, environmental monitoring, object detection and tracking.

Applications of AI in Social-Media: Content recommendation, chatbots and customer support, language translation, user verification, data security and privacy.

Applications of AI in Entertainment: Video and audio editing, visual effects, character animation, virtual reality (VR) and augmented reality (AR).

Applications of AI in Agriculture: Crop monitoring and disease detection, weather forecasting and risk management, automated farming equipment, soil analysis, crop yield prediction.

Applications of AI in E-Commerce: Personalized recommendations, chatbots and virtual assistants, price optimization, analysis of product reviews.

Main Approaches to Artificial Intelligence

Acting Humanly: This approach builds machines that act like humans. The aim is to create a machine that can talk and respond like a person. If people cannot tell whether they are speaking to a human or a computer, the machine is considered intelligent. This idea is based on the Turing Test suggested by Alan Turing in 1950. An example is a chatbot that chats in a natural and friendly way.

Thinking Humanly: This approach makes machines think like humans. It studies how the human brain solves problems and makes decisions and then tries to copy that method. An example is a program that solves a math problem by using the same steps a student would use.

Thinking Rationally: This approach teaches machines to use proper logic and reasoning. It follows clear rules step by step to find the correct answer instead of copying human thought. An example is a program that solves a puzzle using logical rules.

Acting Rationally: This approach makes machines act in the best way to achieve their goals. The machine chooses the most suitable action based on what it knows. It does not need to act or think like a human but simply do the right thing at the right time. This is the most widely used approach in modern AI. An example is a self-driving car that looks at the road and traffic and then chooses the safest and fastest route.

Intelligent Agent

An intelligent agent is something that can sense the environment around it and take suitable action.

Sensors: Sensors are devices that help the agent understand its surroundings by collecting information.

- Examples include a camera that senses images, a microphone that senses sound, and a thermometer that senses temperature.

Actuators: Actuators are parts that carry out actions in the real world based on the decisions of the agent.

- Examples include an electric motor that rotates wheels, a robotic arm that picks up objects, and a speaker that produces sound.

Examples of Agents

- A human being can be seen as an agent. Our eyes, ears, nose, tongue, and skin act as sensors, while our hands, legs, and voice act as actuators.
- A robot can also be an agent. It may use cameras, infrared detectors, or microphones as sensors, and motors, robotic arms, or wheels as actuators.
- Software can behave like an agent too. It senses by reading keystrokes, files, or data, and it acts by displaying results, saving files, or sending messages.

Structure of an AI Agent

The structure of an intelligent agent is made up of two main parts: the architecture (the body or hardware) and the agent program (the brain or software).

Agent = Architecture + Agent Program

Architecture: Architecture is the machinery or platform on which the agent runs. It provides the hardware support for sensors, actuators, and processing. This could be a computer, a robot, a mobile device, or any other hardware system.

- Example: A robot's architecture may include cameras for vision, motors for movement, and a processor for decision-making.

Agent Function: Agent function is like a rule or mapping that tells the agent what action to take for every possible percept (input it receives from the environment).

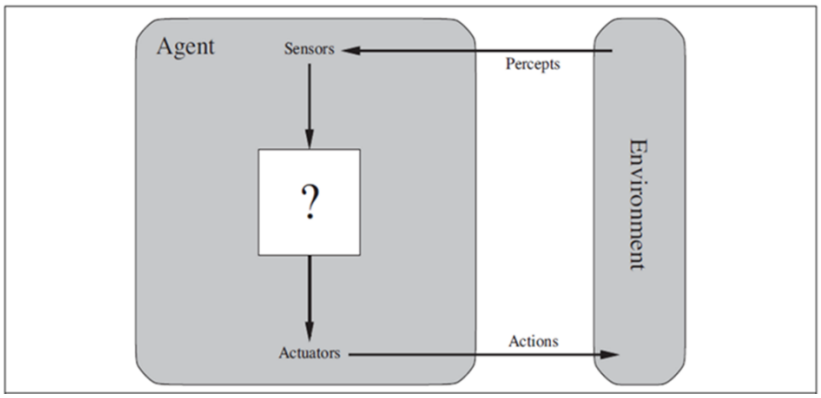
- Example: If a vacuum cleaner senses "dirt," the agent function maps it to the action "suck."

Agent Program: Agent program is the actual implementation of the agent function. It is a set of instructions or code that runs on the architecture and decides the actions to be taken.

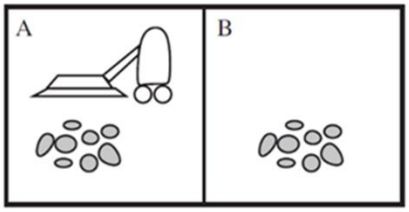
- Example: In a vacuum robot, the program running inside its processor decides when to move, when to clean, and how to act depending on what it senses.

Environment

- The environment is everything around an agent that affects how it works. It is not part of the agent itself, but it provides the things that the agent can sense and react to.
- For a human agent, the environment is the world around us such as roads, buildings, and people.
- For a robotic agent, the environment could be a room, a factory floor, or any physical space where it operates.
- For a software agent, the environment is the computer system, files, internet, or network in which it runs.
- In simple terms, the environment is the "world" in which the agent lives and interacts.



Example for Agents and Environments: A Robot Vacuum-cleaner



- The vacuum-cleaner world has two rooms, A and B. The agent (vacuum) can sense two things: the room it is currently in and whether that room is clean or dirty. Based on this, it decides what to do next.
- The agent has a small set of possible actions: it can suck dirt if the current room is dirty, or it can move left or right to go to the other room if the current one is already clean. In some cases, it could also stay still, but usually, it keeps moving to check.
- The rule it follows is very simple: if the room it is in is dirty, it cleans it by sucking. If the room is clean, instead of staying there, it moves to the other room to check if that one needs cleaning.
- In this way, the agent will keep switching between A and B, sucking whenever it finds dirt and moving when it does not. This ensures both rooms are kept clean without the agent needing to know the whole history of the world.

Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
⋮	⋮

Properties of Task Environments (Nature of Environments)

Fully Observable vs Partially Observable

- If the agent's sensors can see and understand the entire environment at every moment, then the environment is fully observable.
- If the agent can only see part of the environment, then it is partially observable.
- Example: A chess game is fully observable because the agent can see the whole board. A vacuum cleaner is partially observable if it can only sense dirt in its current square but not in others.

Deterministic vs Stochastic

- In a deterministic environment, the next state is completely decided by the current state and the agent's action.
- In a stochastic environment, the next state has some randomness and cannot be predicted with certainty.
- Example: Solving a crossword puzzle is deterministic, because the result is fixed if the right word is chosen. Driving a taxi is stochastic, because traffic and pedestrian behavior are unpredictable.

Single-Agent vs Multi-Agent

- If the agent is the only one operating in the environment, it is a single-agent environment.
- If other agents are also present, then it is a multi-agent environment. These agents may compete or cooperate.
- Example: Solving a math problem alone is single-agent. Playing chess is multi-agent, since two players (agents) are involved.

Static vs Dynamic

- If the environment does not change while the agent is deciding, it is static.
- If the environment keeps changing on its own while the agent is thinking, it is dynamic.
- Example: A crossword puzzle is static because the puzzle does not change. Driving a taxi is dynamic because people, vehicles, and signals change constantly.

Episodic vs Sequential

- In episodic environments, the agent's work is divided into separate episodes where one action does not depend on another.
- In sequential environments, each decision can affect future decisions, so planning is required.
- Example: Classifying images is episodic since each image is independent. Playing chess is sequential because one move affects future moves.

Discrete vs Continuous

- In a discrete environment, there are a limited number of actions or states.
- In a continuous environment, the actions and states can take on a wide range of values that change smoothly.
- Example: Playing chess is discrete because only certain legal moves are possible. Driving a car is continuous because values like speed, steering, and position change smoothly.

Known vs Unknown

- In a known environment, the agent already knows the rules of how actions affect the environment.
- In an unknown environment, the agent must learn the rules through experience.

- Example: Solving a math equation is known because rules are clear. Playing a new game without instructions is unknown.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Examples of Task Environments

Rational Agent

- A rational agent is an intelligent agent that makes logical decisions and tries to do the right thing to reach its goal.
- It does not act randomly but selects the best action that can lead to success. In artificial intelligence, we aim to create such rational agents that can work effectively in different real-world situations like self-driving cars or medical systems.
- The rationality of an agent is judged by how well it performs according to a performance measure. For example, a vacuum cleaner is rational if it keeps the rooms clean, while a self-driving car is rational if it drives safely and reaches the destination on time.
- Rationality also depends on how much the agent knows about its environment, because more knowledge allows it to make better decisions. It also depends on whether the agent can pick the best possible action in a given situation.
- Finally, rationality is based on the whole sequence of inputs or percepts the agent has received, not just the most recent one.

PEAS

PEAS is a framework in AI that explains the main parts of an intelligent agent or system. It helps to define what the system should achieve, where it works, how it acts, and how it senses information.

PEAS stands for:

- P – Performance measure which tells how success is judged.
- E – Environment which is the place where the agent works.
- A – Actuators which are the parts that carry out actions.
- S – Sensors which are the parts that collect information.

Example: Self-Driving Car

- Performance measure includes safety, comfort, following traffic rules, and reaching the destination on time.
- Environment is the real-world roads with pedestrians, other vehicles, traffic lights, and signs.
- Actuators are the steering, brakes, accelerator, horn, and signals.
- Sensors are cameras, GPS, speedometer, sonar, and other devices to understand the surroundings.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

PEAS Examples

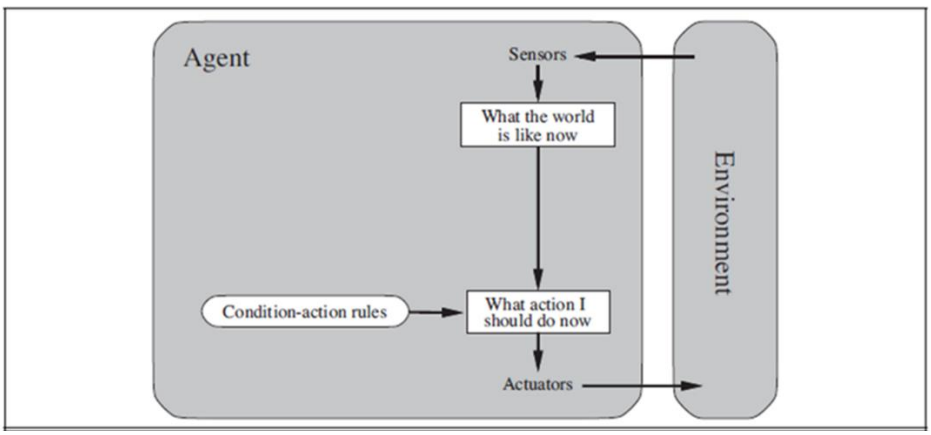
Types of AI Agents/Agent programs

Simple Reflex Agent

- A simple reflex agent is the most basic type of intelligent agent. It makes decisions only on the basis of the current percept it receives and does not remember anything from the past. This means it looks at the situation right now and immediately chooses an action, ignoring all history.
- Such agents work properly only in a fully observable environment, where sensors can see the complete situation.
- The working of a simple reflex agent is based on condition–action rules. For every condition it senses, it has a direct action to perform.
- Example: A vacuum cleaner agent may follow the rule – if there is dirt, then suck. It does not plan ahead or consider other possibilities.

Limitations of Simple Reflex Agents:

- They have very limited intelligence as they cannot consider hidden or unseen parts of the environment.
- They fail in environments that change or are not fully observable.
- The set of condition–action rules can grow too large for complex tasks.
- They are not adaptive, meaning they cannot improve or learn when the environment changes.

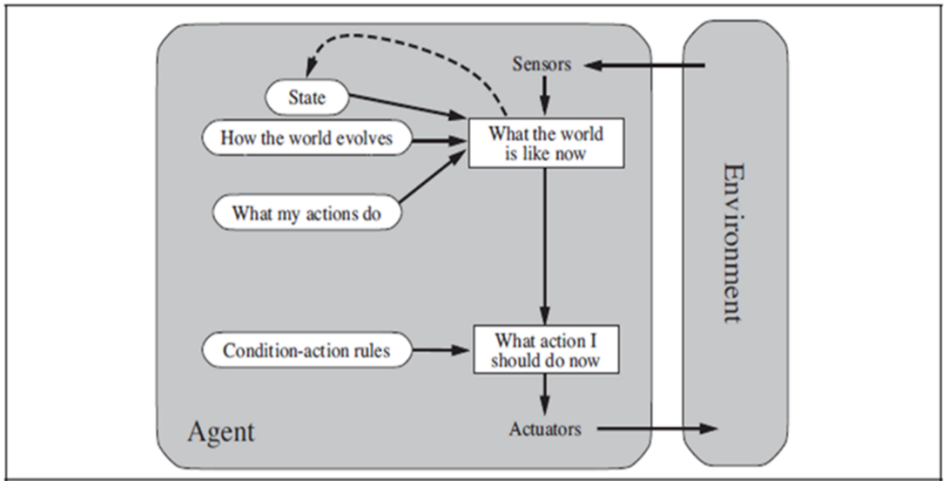


Model-based reflex Agent

- A model-based reflex agent is more advanced than a simple reflex agent because it can work in partially observable environments. Unlike a simple reflex agent, it does not rely only on the current percept but also uses information from the past.
- To achieve this, the agent maintains an internal state that stores details about the part of the world it cannot see directly. This internal state is updated using the agent's percept history. The agent also uses a model, which is knowledge about how the world works and how things change over time.

Example: Consider a vacuum cleaner agent in a house with two rooms, Room A and Room B. The agent can only sense the room it is currently in.

  - If it is in Room A and detects dirt, it cleans it.
  - Later, it moves to Room B and cleans it if dirt is detected.
  - When it goes back to Room A, it remembers from its internal state that Room A was already cleaned. It does not waste time cleaning it again.
- This ability to use memory and knowledge makes the agent more efficient and intelligent than a simple reflex agent. It can handle more complex situations because it relies not just on immediate sensor data but also on stored knowledge of the world.



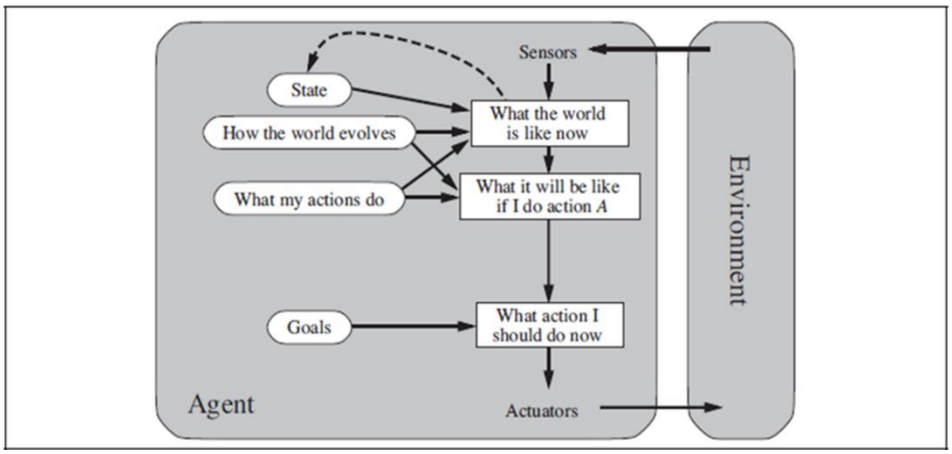
Goal-Based Agents

- A goal-based agent is more advanced than a model-based reflex agent because knowing only the current state of the environment is not always enough to decide what to do. The agent also needs to know its goal, which describes the desired situation it wants to achieve.
- The agent uses its model of the world (knowledge of how things work and how actions affect the environment) together with its goals to make decisions. Instead of reacting only to the current situation, it chooses actions that will eventually lead it toward achieving its goal.
- Example: Consider a self-driving car. Its goal is to reach a destination safely. Just knowing the current traffic light or road condition is not enough. The car must plan a sequence of actions such as turning left, avoiding traffic jams, or slowing down for pedestrians. It may even compare different possible routes before choosing the best one.
- This ability to plan and evaluate future possibilities makes the agent proactive, not just reactive.

Advantages

- Goal-based agents are more powerful and flexible than simple reflex agents or model-based reflex agents because they can:
  - Plan ahead instead of just reacting immediately.

- Take actions that help achieve long-term objectives.
- Adapt better to complex and changing environments.



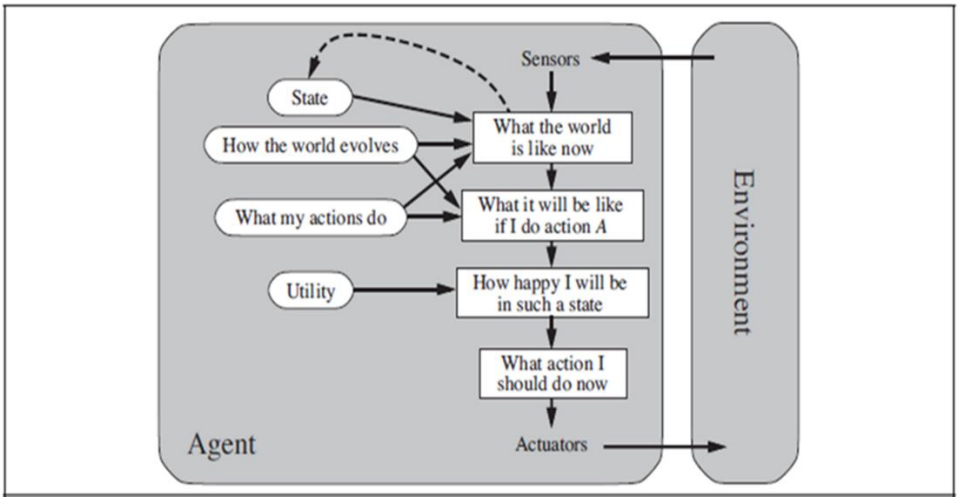
Utility-Based Agent

- A utility-based agent is similar to a goal-based agent but smarter because it does not only check whether the goal is achieved, but also evaluates how good each possible outcome is.
- The agent uses a utility function, which assigns a numerical score to each state, showing how desirable or beneficial that state is. This allows the agent to compare different outcomes and choose the one with the highest satisfaction or benefit.

Example: Consider a self-driving car. Its goal is to reach the destination safely. A goal-based agent would focus only on reaching the destination. A utility-based agent, however, considers multiple possible routes:

- One route may be the shortest but congested with traffic.
- Another route may be longer but smoother and safer.

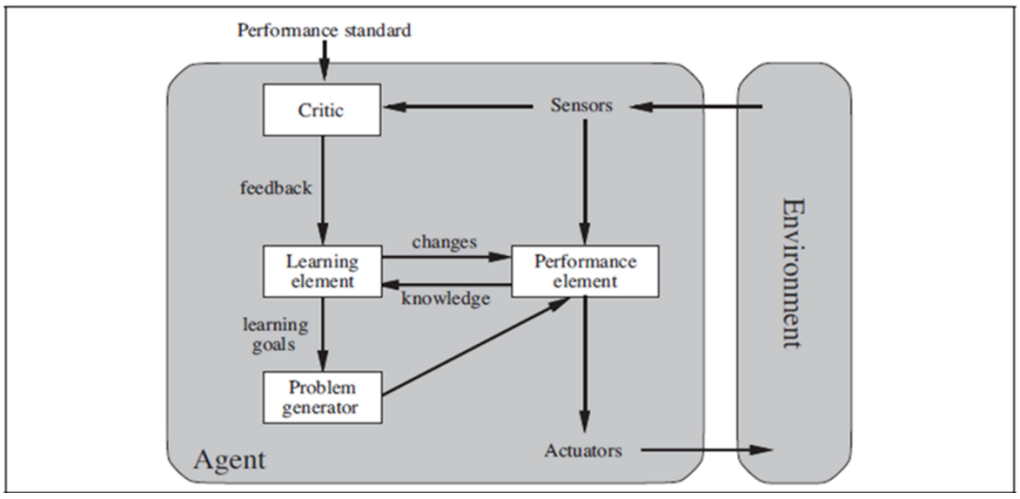
- The utility-based agent assigns scores to each route (considering factors like comfort, safety, and time) and chooses the one with the best overall score.
- This makes utility-based agents more flexible and intelligent, as they can handle situations with multiple goals or alternatives and still pick the best option.



Learning Agents

- A learning agent is an agent that can improve itself by learning from past experiences. Initially, it has only basic knowledge, but as it interacts with the environment, it adapts and performs better over time. It learns what works well and what does not, making it more intelligent with experience.

- A learning agent has four main parts:
  - Performance Element – Takes actions in the environment based on current knowledge.
  - Critic – Evaluates the actions by comparing them to a performance standard and gives feedback on whether they were good or bad.
  - Learning Element – Uses feedback from the critic to improve the agent's knowledge for better future performance.
  - Problem Generator – Suggests new actions or experiments so the agent can gain new experiences instead of repeating the same behavior.
- Example: Consider a self-driving car:
  - The performance element decides when to stop, turn, or accelerate.
  - The critic observes whether the car drives safely or makes mistakes, like taking a wrong turn.
  - The learning element updates the car's knowledge to avoid repeating mistakes.
  - The problem generator encourages the car to try new conditions, like night driving or heavy rain, so it can handle them better in the future.
- Over time, the self-driving car becomes smarter, safer, and more reliable because it continuously learns from its past actions and experiences.



Evolution of AI

- 1943 – First model of artificial neurons developed by McCulloch and Pitts.
- 1950 – Alan Turing proposed the Turing Test to evaluate machine intelligence.
- 1956 – The term "Artificial Intelligence" was coined by John McCarthy at the Dartmouth Conference; AI became an academic field.
- 1960s – Early AI programs focused on solving math and logic problems; first chatbot ELIZA created (1966).
- 1970s – Development of robots; Japan built WABOT-1, the first humanoid robot (1972).
- 1980s – Growth of Expert Systems that mimicked human decision-making; first AI conference by AAAI held (1980).
- 1990s – AI made progress in games; IBM's Deep Blue defeated world chess champion Garry Kasparov (1997).
- 2000s – AI entered the business world; companies like Facebook, Twitter, and Netflix began using AI.
- 2010s – Rapid rise of machine learning, deep learning, and big data.
  - 2012 – Google launched Google Now, a predictive assistant.
  - 2018 – IBM's Project Debater debated with human experts.
- 2020s – Modern AI
  - 2020 – AI used widely in healthcare and COVID-19 research.

    - 2021 – OpenAI's DALL·E and CLIP demonstrated AI can create and understand images from text.
    - 2022 – ChatGPT launched, bringing Generative AI to the mainstream.
    - 2023 – GPT-4 released with advanced reasoning, language, and creativity.
- Now – AI powers self-driving cars, robotics, voice assistants, healthcare, education, finance, and many other areas.

Foundations of AI

Philosophy: Philosophy studies thinking, knowledge, and ethics. It asks whether machines can think like humans or should remain just tools. Philosophy teaches that AI must be fair, safe, and not harmful. It also explores consciousness and free will, raising questions about whether AI can ever be truly "alive."

Mathematics: Mathematics is the language of AI and provides rules to solve problems. Logic helps AI reason step by step. Probability and statistics allow AI to handle uncertainty and make predictions. Linear algebra is used to store and process data such as images, videos, and text. Calculus helps AI models learn and improve by optimizing performance.

Economics: Economics is about making smart decisions with limited resources. Utility helps AI choose the most beneficial option. Game theory shows how AIs or humans can compete or cooperate strategically. Economics helps AI optimize outcomes efficiently. Example: Companies use auction models to sell ad spaces and maximize profit.

Neuroscience: Neuroscience studies the brain and nervous system, showing how humans learn, remember, and pay attention. It inspired artificial neural networks, enabling AI to perform brain-like tasks such as object recognition, speech understanding, and decision-making. Example: The human visual system inspired computer vision in AI.

Psychology: Psychology studies how people think, learn, and behave. It explains memory, problem-solving, and attention, which inspire AI design. It helps AI become more user-friendly and interact naturally with humans. It also highlights human mistakes and biases that AI should avoid. Example: Learning apps use step-by-step practice based on psychology methods.

Computer Engineering: Computer engineering provides the hardware and software needed for AI. Hardware includes processors, GPUs, sensors, and robots, while software includes programming languages and operating systems. It ensures AI runs fast, securely, and reliably, even with big data. Example: Smartphones use computer engineering to run assistants like Siri.

Control Theory: Control theory studies feedback systems that adjust automatically. It works in a cycle: sense → act → check → correct. This keeps machines stable and prevents failure. Example: A thermostat controls room temperature by switching heating on or off.

Linguistics: Linguistics studies language, grammar, meaning, and context. It helps AI understand speech and text accurately, solving problems like ambiguity where the same word has different meanings. Example: The word "bank" can mean a riverbank or a financial bank. Linguistics is used in chatbots, translation apps, and voice assistants.

How the components of agent programs work?

Three different ways in which an AI agent can represent its environment: **Atomic, Factored, and Structured representations**.

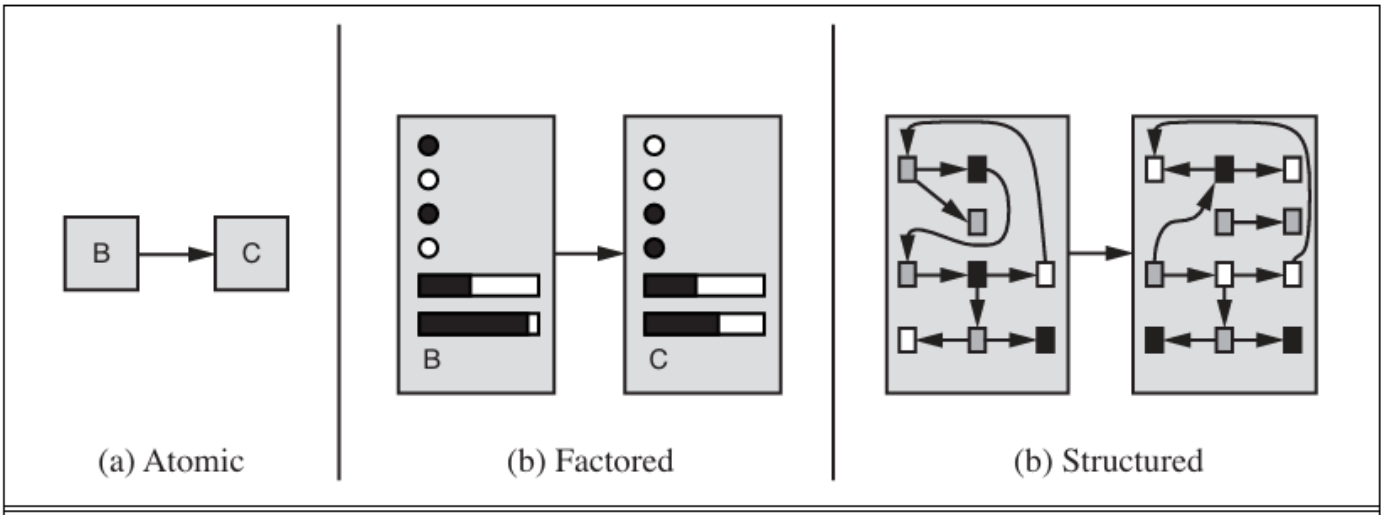


Figure 2.16 Three ways to represent states and the transitions between them. (a) Atomic representation: a state (such as B or C) is a black box with no internal structure; (b) Factored representation: a state consists of a vector of attribute values; values can be Boolean, real-valued, or one of a fixed set of symbols. (c) Structured representation: a state includes objects, each of which may have attributes of its own as well as relationships to other objects.

In an **atomic representation**, each state is treated as a single, indivisible unit or **black box**.

EX: Consider finding a route between cities. The AI may represent each state simply as:

Bengaluru, Mysuru, Chennai.

The AI does not consider additional details about the city. Each state is just a single piece of information.

A **factored representation** divides a state into a fixed set of **variables or attributes**, where each variable has a value.

EX: For a driving agent, the state might be represented as:

Variable	Value
Location	Bengaluru
Fuel level	20 litres
GPS	Active
Oil warning light	Working
Money	₹500

A **structured representation** represents
objects and the relationships between them.
EX: For a autonomous vehicle, there is a cow blocking
its way. Cow is moving towards the farm.
Car is moving towards the farm. Man is moving away from farm and so on…